

Third generation mobile systems

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This paper sets out to review the next generation of mobile systems, to explain the essential differences from the second generation (digital) systems and to look at activities on a world basis. There is currently much activity in this area as the UK, Europe and the rest of the world community have begun to recognise the opportunities of the mobile radio vision.

1. Introduction

This paper discusses third generation mobile systems (TGMS) — the evolution of mobile communications towards the goal of universal personal communications, a range of services that can be anticipated being introduced early in the next century to provide customers with wireless access to the information superhighway and meeting the ‘Martini’ vision of communications with anyone, anywhere and in any medium. This goal is known by a number of acronyms — Future Public Land Mobile Telecommunications Systems (FPLMTS) being the most common at an international level, although this has more recently been replaced by IMT-2000 (International Mobile Telecommunications — 2000, the latter representing both the year of introduction and the frequency, in MHz, of operation). Within Europe the term Universal Mobile Telecommunications System (UMTS) has been applied to the standards development for the next generation system although, more generically, third generation mobile systems or TGMS is in common use; the latter will be used throughout this paper as the general term for future mobile telecommunications systems. UMTS will be reserved for the specific European standards development and FPLMTS for those of the ITU.

It must also be borne in mind that ‘mobile’ does not imply only wireless or cordless terminals, for while these will inevitably represent perhaps the most significant number of users in the next century, there is also a demand for providing mobility on fixed networks enabling customers to access a range of personal services from any (fixed) terminal.

First generation analogue mobile systems are now very well established world-wide but operate under a wide variety of technical standards. Second generation digital systems are becoming well established but again the technical standards differ between continents, notwithstanding the undoubted success of GSM. The goal for third

generation mobile systems, as originally articulated within the ITU in 1986, was a single world-wide technical standard providing a universal service offering, at a cost commensurate with world-scale economies of manufacture. The commitment to a world-wide common air interface standard is now being re-assessed as other technologies, particularly the adaptive multimode terminal, provide an alternative approach to addressing the range of frequencies and potential coding techniques that will offer users services in a wide range of physical environments; the need, however, for a common approach, world-wide, for core network signalling and transmission standards and the development of a broadband ISDN, based on intelligent network standards, remains paramount. However, even with adaptive multimode terminals, there is still a likely need for a common ‘pilot’ channel to communicate the appropriate access standard to the terminal and avoid the user having to understand and select technology types.

Starting with the vision for TGMS, an outline is provided of what TGMS is and why it is needed. Then the key issues that need to be addressed and the way in which we might evolve towards TGMS from present generations of communications networks are considered. European and worldwide activities are then reviewed and the paper concludes with a discussion of the challenges that face the telecommunications industry if the third generation mobile system is to be realised.

2. Vision

There is no shortage of visions for TGMS. Probably the vision for mobile radio systems is that of more wireless than fixed terminals connected to the telecommunications networks world-wide. The market research from the European RACE programme, looking at third generation systems, suggested, in 1990, that within ten years 50% of all

calls would originate or terminate on a mobile terminal, provided that they offered toll-quality speech at PSTN prices. The European Green Paper on Mobile Communications [1] published in April 1994 predicted that within the next two decades, the market for mobile terminals within Europe would reach 80% of the population. If this proves correct there will be some 280 million mobile terminals compared with, perhaps, 180 million fixed connections. There are around 150 million fixed connections within Europe at present and the growth rate for mobile connections now exceeds that for fixed. As there are currently some 25 million mobile connections in Europe, a tenfold increase in these is anticipated over the next 20 years. As shown in Fig 1 the consequences of this growth will be a customer demand for common call set-up and control procedures across both fixed and mobile networks and for an equal quality of service.

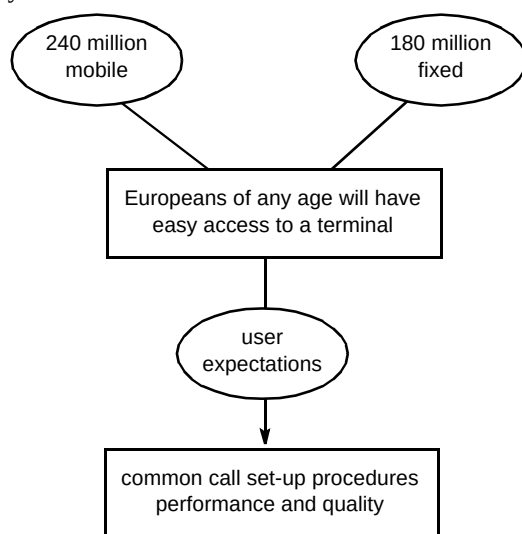


Fig 1 Future demand for common procedures.

Indeed, recognition of this scenario provides the foundation for what is commonly known as fixed-mobile convergence (FMC), in that the prime needs of future telecommunications customers will be for:

- a single number per role (e.g. private and business),
- a single bill,
- a single point of contact for sales and service,
- common operating procedures,
- common call-control procedures.

The impact of this on existing network operators, in terms of both commercial opportunity and network infrastructure to meet the signalling and database load associated with authenticating, tracking and billing such a mobile customer base, cannot be understated.

Nor are such future mobile systems likely to be monopolies under the control of a single network operator or service provider within any one country, let alone within a continent or world-wide; this, in turn, will imply new approaches both to customer service provision and to the development of the regulatory and licensing regimes for telecommunications networks throughout Europe — a topic outside the scope of this paper.

A widely accepted vision for TGMS is that of the universal personal communicator:

- universal — it can deliver services that meet varying communications needs to a user at any time and in any place,
- personal — the services are configured to meet the needs of that user from a small terminal that is carried in the pocket,
- communicator — the terminal is more than just telephony, encompassing multimedia, data, paging and video communications.

The service will include both public and private access — the former being that currently associated with cellular systems and wireless local loop, the latter with cordless systems. Private would include business premises access from a wireless PBX and customer premises access from a home base station, or private mobile radio to major companies.

Most importantly, the system is expected to provide high-quality services at a cost that is affordable to the mass market, both in terms of terminal cost and usage charges.

To put the service in context it is seen as combining the attributes of a current cordless handset in terms of size, weight and battery life, with the functionality and ease of use of the now-emerging personal digital assistant products to provide the user with access to any information, including moving image, during a call.

3. What is third generation?

How, then, do the aims of TGMS differ from the existing and developing systems? While the precise nature of TGMS is yet to emerge, a useful starting point is found in the recently published UMTS Task Force Report [2] which states the next generation systems will:

- take the personal communications user into the new information society,
- deliver information, pictures and graphics directly to people and provide them with access to the next generation of information-based services,

- move mobile and personal communications forward from the second generation systems that (will) have delivered mass-market low-cost digital phone calls.

Mobile systems, like any other complex development, take some time to move from initial concept through to fruition and place their users' needs (services, features, etc), as a foremost requirement; these, however, tend to be the needs at the time that the development programme is initiated. With sufficient forethought, of course, flexibility will be built in to ensure that future needs can be met; although there is, of course, always some limitation on the amount of flexibility that can be provided.

In cellular system developments, for instance, the key drive to maximise the number of users in a minimum of frequency spectrum leads to the use of low-rate speech coders and the provision of low-rate data services.

Cordless developments have taken a different route in that although their primary drive was similar to public cellular — to maximise the number of users on a limited frequency spectrum — they have achieved this by limiting the power and range of the terminals rather than limiting the bandwidth of the services.

Private mobile radio (PMR) systems have traditionally adopted a wide-coverage, lower-quality communications route in maximising user capacity on limited spectrum.

Each of these approaches is equally valid, and is tailored to meet the specific need of the market addressed; however, with present-generation systems, cellular, cordless and private mobile radio systems are not compatible with each other.

The major objective of TGMS is seeking convergence across all market sectors, including satellite.

3.1 TGMS user features

Customers will be looking for an affordable, high-quality communications system. Quality will be particularly important as customer expectation grows year by year with their increased exposure to improved quality audio and video systems. What users may have deemed an acceptable quality two or more years ago will no longer be sufficient. Digital transmission has improved the quality of wired communications — customers will want this to continue in the mobile domain. As another leap in the demand for quality can be anticipated by the year 2000, TGMS needs to address this from the outset.

The services will have to be both those which are commonplace today and those that could be expected to be readily available in the year 2000, such as (mobile)

videophone and high-speed data applications, as well as general-purpose telephony. Both narrow and broadband ISDN-based services will become more available and be widely used and customers could also be expected to manage and customise their services to their own particular needs through the broadening use of intelligent networks.

Taking these broad services and features, the user might expect to see:

- broadband and multimedia capability,
- flexible bandwidth-on-demand allocation,
- speech quality comparable with fixed network,
- packet data capabilities,
- global availability and global roaming,
- terrestrial and satellite capability in one system family,
- mass-market and niche-market products in one system.

The key differentiator over second generation systems is that of broadband service delivery, typically thought of as 2 Mbit/s access, coupled with bandwidth-on-demand allocation; this latter may be used to download a large file during a voice conversation or to secure (and pay for) a higher quality speech channel. It is also recognised that for wide-area mobile applications the access rate may well fall to around 144 kbit/s. The rest of the above expectations address the demand for a service which is universally available and encompasses both public and private systems.

If these numerous user services are to be supported, in a quality way, in diverse environments, with a limited frequency spectrum, and all of this at a low cost, what are the important issues to keep in mind during the third generation systems developments?

3.2 TGMS key system features

If these needs are now translated into the high-level system requirements, TGMS will provide service to:

- remote areas,
- rural areas,
- residential areas,
- indoor offices,
- vehicles and pedestrians.

Further, to exploit fully the spectrum allocation, while maximising capacity, a flexible cell architecture will be needed, providing a mixture of overlay, micro- and pico-cells; since these must also encompass a mixed public and

private domain, frequency planning will need particular attention, favouring those approaches that provide adaptive channel allocation. Therefore TGMS is likely to employ a layered radio-cell architecture:

- with mega-cells using satellites for remote areas ,
- with macro-cells for wide area (ubiquitous) coverage,
- with micro-cells supporting high density urban areas,
- with pico-cells for indoor use,
- from a mixture of both public and private networks,
- with a number of operators,
- with networks that must locate and track the user during a call and bill correctly.

Figure 2, taken from the Task Force Report [2], summarises the proposal.

Regulatory and commercial pressures will demand open network provision (ONP) and in this it is expected there will be a number of stakeholders ranging from application providers, service providers and network operators, through to the end users themselves, all of whom will have a major role to play in ensuring that TGMS is the quality system that the end user demands as well as being a commercial success.

Nor can radio spectrum be magically created overnight for TGMS — there will still be a constraint on fitting the maximum number of users into the minimum spectrum. Some 230 MHz of radio spectrum has been made available at WARC 92 for FPLMTS and there are ongoing discussions as to its detailed use. It has also been recognised [2] that the allocation is unlikely to prove sufficient for the envisaged market.

Finally, the networks themselves — these networks are expected to become increasingly ‘intelligent’, offering more services in a high-quality manner and at competitive prices. There is likely to be increasing competition with value-added service providers, as well as network operators, providing users with more choice in their communications needs. In summary TGMS will provide:

- multimedia communications,
- from multi-cell radio access networks,
- in a multi-operator/service provider environment.

3.3 TGMS system development

TGMS will need to support numerous user services, in a quality way, in diverse environments, with a limited frequency spectrum, and all of this at a low cost. What are

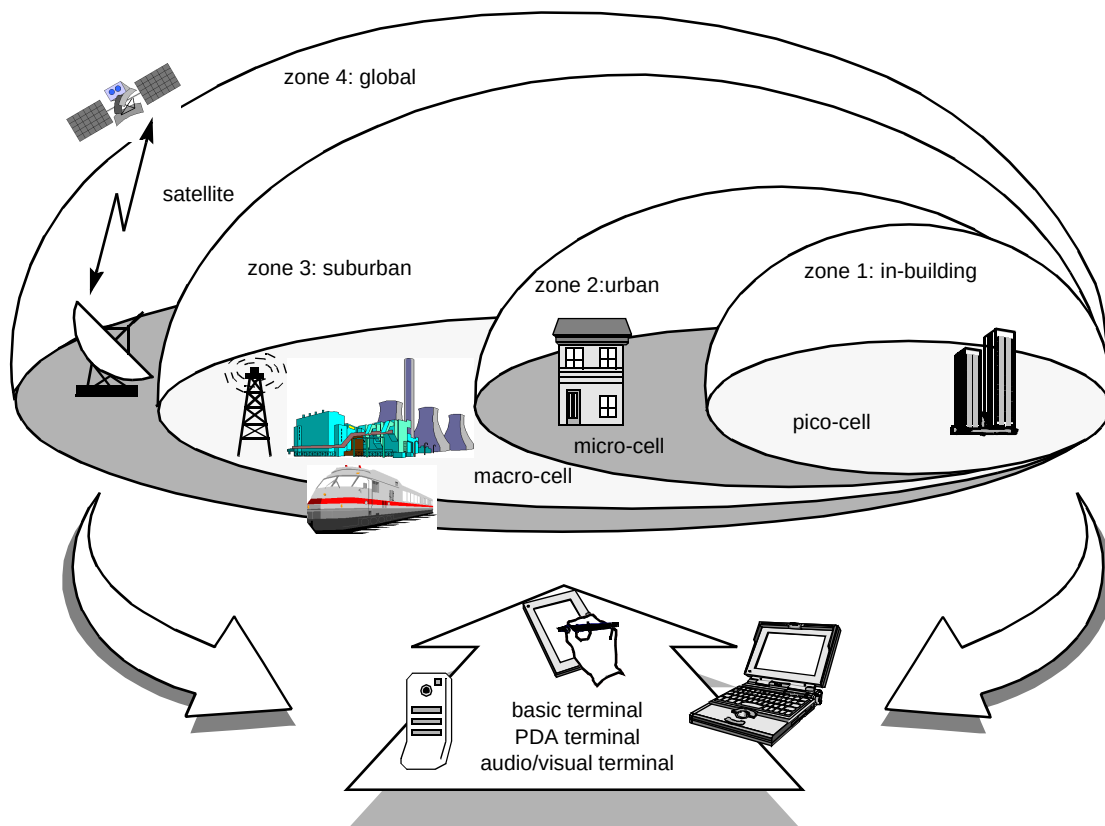


Fig 2 Integration with the fixed network.

the important issues that must underpin third generation system developments?

The key need for TGMS is flexibility. Flexibility in the air interface or access technology will be required to support bandwidth-on-demand services in which the user requests (and pays for) the actual bandwidth that a particular service requires; this could be dependent on the time of day and an example would be variable-rate speech codecs — a user who accepts lower quality speech, such as for a PMR application, will require limited bandwidth, whereas a user who demands 'hi-fi' quality will be allocated, and charged for, a higher bandwidth channel.

The need is to provide the appropriate bandwidth for the specified service rather than, as at present, a given (and usually minimal) bandwidth irrespective of the user service being supported.

The network developments must allow for value-added service providers, as well as network providers and, perhaps, even radio access providers — all to be supported on a common architecture. Third generation developments should not preclude any regulatory or national implementation, nor should they impose any specific implementation on operators, or service providers.

The provision of a dense cell architecture, to support the envisaged level of traffic, will result in a high network infrastructure cost. There will thus be a major drive to integrate both the fixed and mobile network portion of the system to allow operators to minimise duplication of equipment and provide an efficient, low-cost network capable of providing services in an integrated and seamless manner, at an affordable cost.

A flexible service provision capability is a further requirement in providing a rich set of services. To that end TGMS must aim for integration with intelligent network developments as highlighted by Cullen and Lobley [3]. Integration with IN is critical in minimising unnecessary network provision and is an essential step towards providing a management capability that will address the service interactions that can occur between mobility and fixed user services.

4. Standards development

Far from the unified and integrated vision of third generation mobile systems, the current position in the UK is probably typical for most countries where separate paging, cordless, cellular, PMR and mobile satellite services have developed over the past ten or more years. While these systems are presently emerging from analogue to digital implementation, the TGMS goal is an expectation that these will all combine early in the next century.

Studies in the ITU, which were initiated in 1986, have identified a range of radio interfaces for FPLMTS as shown in Figs 3 and 4 and which address those areas currently served by independent cordless, cellular and satellite systems.

The ability of a common radio interface to meet the requirements identified for TGMS is a critical consideration; however, if a common interface at a physical radio level, i.e. modulation and channel coding, does not prove possible, what remains as essential, if the goals of universal service to users is to be achieved, is the provision of common procedures for mobility management and user service control.

The critical issues for the development of the air interface are, therefore:

- preferably a single standard for the basic radio transmission and coding, but, if this does not prove possible, sufficient commonality in the different interface specifications to allow for intelligent terminals to be produced that can adapt to more than one standard and minimise the implementation cost,
- definitely a single standard for the basic mobility and user control mechanisms — this is essential in enabling the terminal to access the network, make calls, perform a handover, and update locations.

Given these objectives and the broadband nature of the radio interface, it is difficult to envisage a specific evolutionary path from any existing cellular or cordless technology; this suggests that the access technology for third generation mobile systems will be revolutionary. Possible candidates are CDMA or advanced TDMA structures which have been addressed in the RACE mobile projects CODIT and ATDMA.

The critical issues for development of the network components of TGMS are:

- the provision of flexibility in service control and provision,
- the integration of fixed and mobile networks,
- minimising infrastructure cost while providing dense network coverage,
- integration of mobility with network management and service creation capabilities.

All of these can be addressed by steering intelligent network developments to encompass the specific mobility needs, both terminal and user, of TGMS. Progress towards this has been made within Europe through the RACE MONET collaborative project and the issue is recognised in the European Telecommunications Standards Institute (ETSI) committees SMG, SPS and NA6 for UMTS. Such an approach is also being adopted and progressed within the

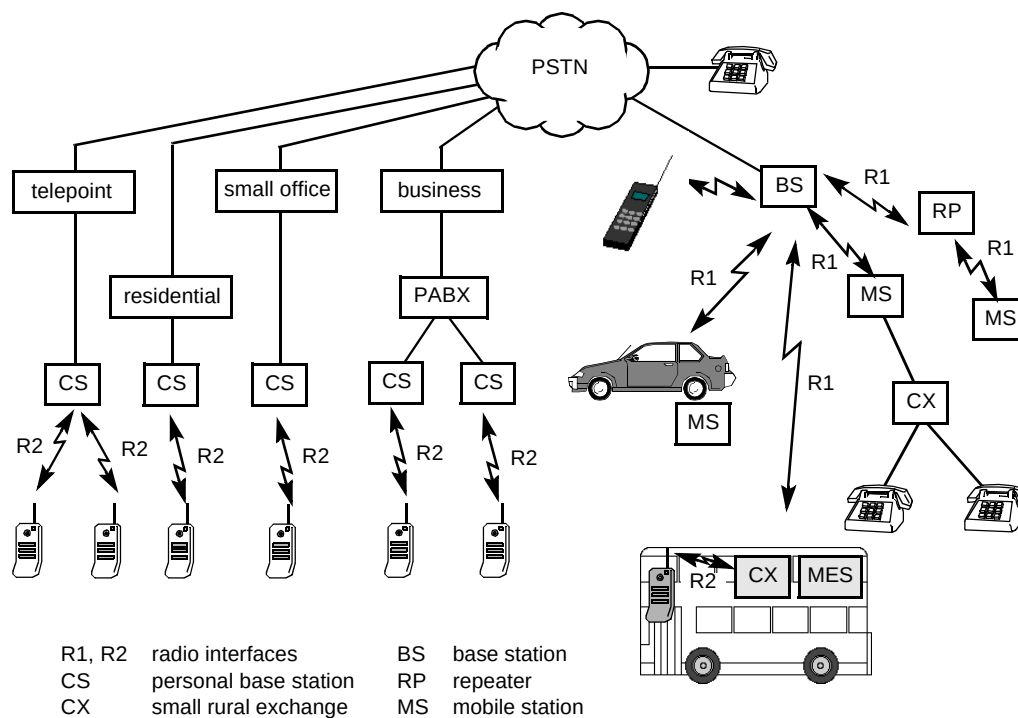


Fig 3 FPLMTS — terrestrial component.

ITU/TSS developing the FPLMTS system requirements. While recognition has been made that the FPLMTS network is an evolution of IN, the impact of this development on IN should not be underestimated. There are considerable developments needed before it can become a reality.

While at an international level the work under the ITU is directed at the FPLMTS standard, on a world basis, more immediate effort is being expended on present and intermediate stage systems. While Europe recognises and supports the UMTS development, CT2/DECT cordless

systems and cellular GSM, particularly, are attracting the greatest development activity (see Holley [4], and Merrett and Buttery [5]). In the United States it is personal communication services (PCS), rather than third generation systems, that are receiving the attention and in Japan effort is directed towards the Japanese 'Digital Cellular' and 'Personal Handyphone' standards, both of which use second generation technology.

It is important to keep on the TGMS vision in sight, if the future service offerings are not to be compromised by

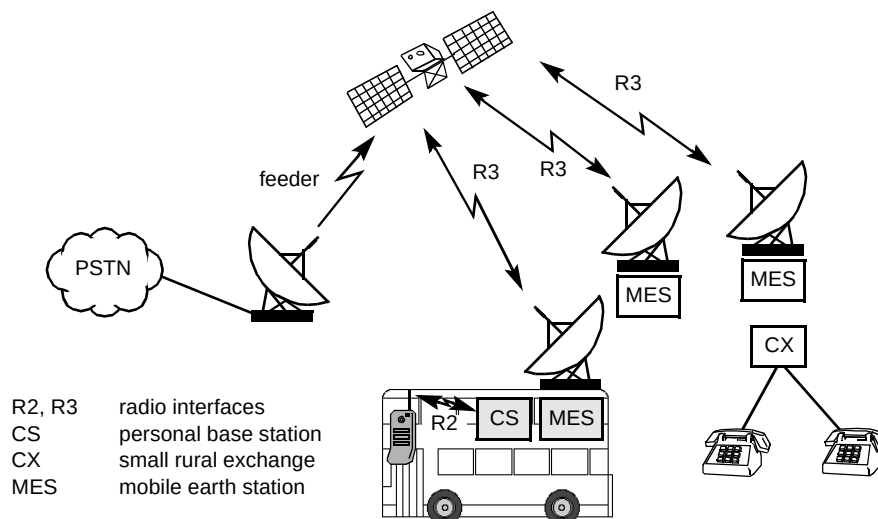


Fig 4 FPLMTS — satellite component.

the evolution of existing second generation standards within Europe, Japan and the USA.

5. Intelligent multimode terminal

The various radio interfaces that may be needed in a third generation mobile system have already been highlighted, but, at this stage, and until further research has been undertaken, it is not possible to decide which approach is optimum for any radio interface or whether more than one type of interface will be appropriate.

The TGMS mobile terminal may thus be required to adapt its multiple access method, modulation, source and channel coding depending on the type of service required and the mobile's location. Such a mobile, the intelligent multimode terminal, was proposed by Steele [6], the transmitter section of which is shown in Fig 5. While the complexity of such an approach far exceeds that of any second generation terminal and not every user will require the full range of potential system interfaces, such a terminal is likely to come into being over the next decade. It is envisaged that the terminal would interact with a particular base station's control channel to agree and set its appropriate radio, modulation and coding parameters, requiring only a minimum of international standardisation and a common design would benefit from the economies of scale afforded by world-wide manufacture.

6. Conclusions

This paper has addressed the vision for third generation mobile systems identifying their broadband, flexible bandwidth-on-demand capability as the prime service dif-

ferentiator over present second generation mobile systems. Global roaming is a further objective for TGMS, but this may well be achieved through the use of adaptive multimode terminals rather than through a common global standard for the radio interface.

It is likely that the air interface will be a revolutionary development but, more importantly, the supporting network infrastructure should be an evolutionary convergence of GSM, DECT, B-ISDN and IN.

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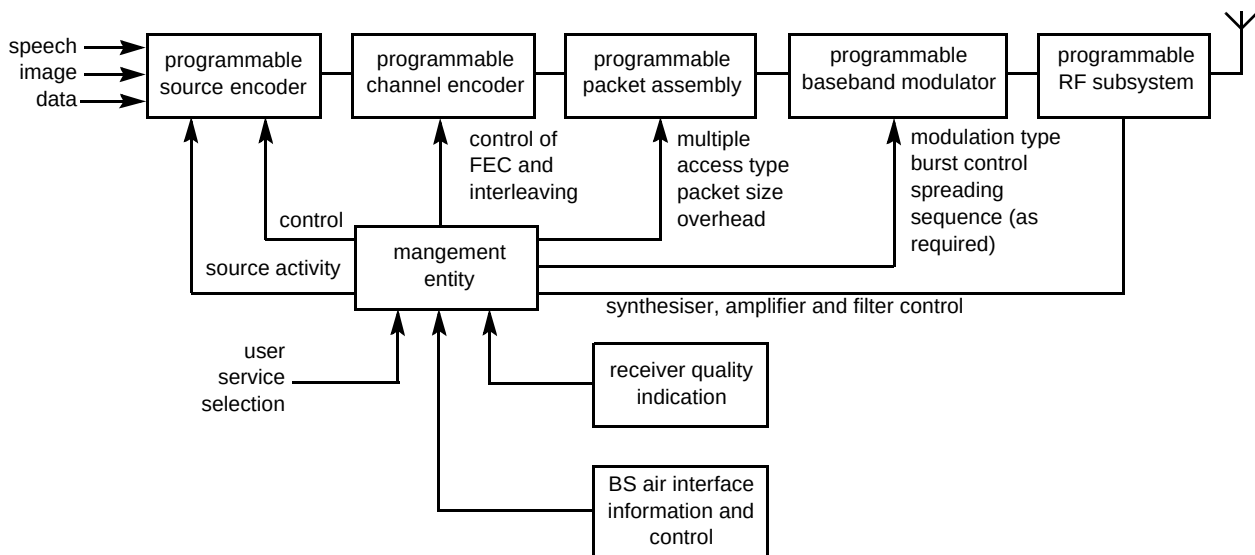


Fig 5 Intelligent multimode transmitter.



Ian Groves was recently appointed Principal Advisor, Mobile Systems at BT Laboratories, having previously headed the Mobile and Radio Networks Division. He joined the then Post Office Research Station in 1968 after graduating from University College, London, and spent the initial part of his career with the Trunk Circular Waveguide Division, developing millimetre-wave oscillators and modulators. He obtained a PhD in 1975 for his work on IMPATT diode oscillators and in 1976 transferred to the Speech Terminals Division, working initially on audio teleconference systems and co-filing three patents. In

1980 he was appointed leader of the development teams responsible for the first electronic telephone instruments developed specifically for the BT product range and held personal responsibility for speech transmission issues associated with customer premises equipment. From July 1986 he headed the Telephony and Data Products Division until his transfer in October 1989 to head what was then the Mobile Systems Division. He is a fellow of the IEE and member of the IEEE.



Alan Clapton started his career with BT in 1967 as an apprentice at the high frequency radio station at Leaffield near Oxford, and subsequently graduated with a BSc honours degree in Electrical Engineering.

Following a period of work in aeronautical and maritime on radio development systems, a move to BT Laboratories opened up work on GSM service and network developments. Recent work has included assisting Cellnet in developing the GSM data and value added services as well as the introduction of intelligent network capabilities. In parallel to this he has led the BT team involved in the

UMTS RACE MONET project, which has focused on IN, broadband and mobile integration.